SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING COURSE
CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I P. Sai Bhargav (192411049) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: P.Sai Bhargav

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3	Level 1 – Beginnin

				Marks)	g (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with	Incorrect or irrelevant answer

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① Responsive web Design:

Responsive web design (RWD) is a web development approach that enables a website to automatically adjust its layout, content, Images and navigation according to the screen size and device the being used.

Importance of Responsive web design.

* It Improves user experience by making websites easy to read and use on all devices
* It avoids the need to create separate websites for mobile and desktop users.

Key css Techniques used for Responsiveness:

① media queries: Media queries apply different css style based on the device's screensize.

```
@media (max-width:768px){
  .container{
    flex-direction: column;
}}
```

② flexible widths:

Relative units such as percentage (%), vw and vh allow elements to resize according to the available screen space.

```
.container{
  width: 90%;
```

margin: auto;

}

③ Responsive Images: Images can be made responsive by setting their maximum width to 100%.

```
img{
    max-width: 100%;
    height: auto;
}
```

④ Relative units: units such as rem, em, %, vw, and vh create flexible text and layouts.

```
h1{
front-size: 2rem;
}
```

⑤ flexible layout systems: css flexbox and css grid help arrange webpage elements dynamically and follow layouts to adapt to screen sizes.

② client side validation:

client-side validation is the process of checking user input in the web browser before the form is submitted to the server.

→ All required fields are filled.

→ the email address is in a valid format.

→ the password meets the required length.

→ the entered data follows the expected format.

Purpose and Importance:

* Improves user experience by displaying errors.
* Reduces incorrect (or) incomplete form submissions.
* prevents unnecessary requests form being sent to the server.
* Reduces server processing for simple input errors.

Javascript event handling in validation.

Javascript event handling allows a program to respond when a user performers an action. common events used inform validation include:

- submit - occurs when the users submits. the form
- blur - occurs when the user leaves an input field.

Role of Regular expressions:

A regular expressions also called Regex. is a pattern used to check whether the input follow a required format

→ Text before @

→ The @ symbol

→ A domain name

→ A valid domain extension.

(3) css Box Model and Responsive layout

when a webpage layout breaks after resizing the browser window, the problem may be caused by fixed widths.

incorrect positioning. large Images, over flowing content (or) missing responsive css rules.

① css Box Model:

The css box model explains how every HTML element is displayed as a rectangular box. each element - contains four main components

* content: the content area contains text, Images, buttons (or) other information.
* padding: padding is the space b/w the content & the border.
* Border: the border surrounds the padding & content.
* Margin: margin is the space outside the border.

```
.box {
   width: 300px;
     padding:20px;
     broder: 5px solid black;
       margin15px;
}
```

By default the actual width is calculated as total width = content width + left & Right Padding + left and right border + left and right margin.

Thefore : total width = 300 + 40 + 10 + 30

= 380 pixels.

using box sizing:

{

box sizing : border-box;

}.

④ document object Model (DOM) & Dynamic webpage updates.

The document object Model is a programming interface that represents HTML webpages as a structured tree of objects each HTML element.

<body>

<h1> welcome </h1>

<p> hello user </p>

</body>

the DOM as structure is represented a

Document

└→ HTML

  └→ head

  └→ Body

    └→ H1

      └→ welcome

    └→ P

      └→ hello user.

Roles of the DOM in web development:

① change text content dynamically
② update Images without reloding the page.
③ chang css styles
④ Add new HTML elements
⑤ Remove existing elements
⑥ display real-time information
⑦ creates interactive forms, menus & Dashboard
⑧ create and show notifications & messages.

Javascript DOM manipulation.

Javascript can acess HTML elements using method such as

1. get element By Id()

const title=
doc.get element by Id ("title");

2. query selector()

This method selections first element that matches a css selector ~~const~~

const button= doc.get query selector ("button");

3. query selector All()

this method selects all elements that match as a css selector.

const cards =
doc.query selector All(".card");